Diagonalization of Matrices

Linear Algebra - 24DS

QUEST - 24DS

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What is Diagonalization?

Definition: A square matrix is diagonalizable if there exists an invertible matrix and a diagonal matrix such that:

$$\boxed{{}^{-} \; =}$$

Equivalently:

$$= \; {}^{-}$$

Where:

- $=$ diagonal matrix of eigenvalues
- $=$ matrix whose columns are eigenvectors

Why diagonalize?

- $= \; {}^{-}$ (easy powers!)
- Simplifies solving differential equations
- Understanding linear transformations

The Diagonalization Theorem

When is a matrix diagonalizable?

An $\times$ matrix is diagonalizable if and only if:

$$\boxed{\text{has linearly independent eigenvectors}}$$

Equivalent conditions:

1. Geometric multiplicity $=$ Algebraic multiplicity for **each** eigenvalue
2. Sum of geometric multiplicities $=$
3. Minimal polynomial factors into distinct linear factors

Not all matrices are diagonalizable!

Case 1: Distinct Real Eigenvalues

Always Diagonalizable!

Key Fact: If has distinct eigenvalues, then is diagonalizable.

Why? Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Example in $\mathbb{R}$:

$$= \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

(Already diagonal! Eigenvalues: 2, 3, 5 — all distinct)

Example 1: Three Distinct Eigenvalues

$$= \begin{bmatrix} \\ \\ \end{bmatrix}$$

Step 1: Find eigenvalues $\det(- \lambda) = $:

$$\det \begin{bmatrix} -\lambda & & \\ & -\lambda & \\ & & -\lambda \end{bmatrix} = (- \lambda)(- \lambda) =$$

Eigenvalues: $\lambda =$ (multiplicity 2), $\lambda =$ (multiplicity 1)

Only 2 distinct eigenvalues! Let's find eigenvectors...

Example 1: Finding Eigenvectors

For $\lambda = :\ - = \begin{bmatrix} - & & \\ & - & \\ & & - \end{bmatrix} \to \begin{bmatrix} & - & - \\ & & \\ & & \end{bmatrix}$ Eigenvectors: $= +$

Two independent eigenvectors:

$$\mathbf{v} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Geometric multiplicity $= 2$ (equals algebraic multiplicity)

For $\lambda = :\ - = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \to$ eigenvector $\mathbf{v} = \begin{bmatrix} \\ - \\ \\ \end{bmatrix}$

$\boxed{\text{3 independent eigenvectors} \Rightarrow \text{Diagonalizable!}}$

$$= \begin{bmatrix} & - \end{bmatrix}, \quad = \begin{bmatrix} & \end{bmatrix}$$

$$= {}^{-}$$

Check: $\mathbf{v} = \mathbf{v}$, $\mathbf{v} = \mathbf{v}$, $\mathbf{v} = \mathbf{v}$

Even with repeated eigenvalues, we had enough eigenvectors!

Case 2: Repeated Eigenvalues

Sometimes diagonalizable, sometimes not

Key: For each eigenvalue λ with algebraic multiplicity :

$$\boxed{\text{Geometric multiplicity} = \dim(\text{eigenspace})}$$

Condition	**Diagonalizable?**
Geo mult $=$ Alg mult for all λ	YES
Geo mult ¡ Alg mult for any λ	NO

Example where it works: Previous matrix ($\lambda =$ had geo mult $= 2$)

Example 2: Repeated Eigenvalues

Diagonalizable despite repetition

$$= \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

Eigenvalues: $\lambda =$ (mult 2), $\lambda =$ (mult 1)

Eigenvectors:

- For $\lambda =$: $(,,)$ and $(,,)$ — geo mult $= 2$
- For $\lambda =$: $(,,)$ — geo mult $= 1$

$\boxed{\text{Diagonalizable!}}$ Already diagonal!

Example 3: NOT Diagonalizable

Defective matrix

$$= \begin{bmatrix} & \\ & \end{bmatrix}$$

Eigenvalue: $\lambda =$ (algebraic multiplicity $= 3$)

Find eigenvectors: $- = \begin{bmatrix} & \\ & \end{bmatrix}$

$$\begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} \\ \end{bmatrix} = \mathbf{0}$$

Equations: $=$, $=$ free Only **one** eigenvector: $\mathbf{v} = (, ,)$

Geometric multiplicity $= 1$ ¡ 3 NOT diagonalizable!

Example 4: Mixed Real Eigenvalues

$$= \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Eigenvalues: $\lambda = , ,$ (all distinct)

Eigenvectors:

- $\lambda = :\ (, ,)$
- $\lambda = :\ (, ,)$ (since $-$ gives $(, ,)$ relation)
- $\lambda = :\ (, ,)$

$\boxed{\text{3 independent eigenvectors Diagonalizable!}}$

$$= \begin{bmatrix} \\ \\ \\ \end{bmatrix}, \quad = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Example 5: Including Zero Eigenvalue

$$= \begin{bmatrix} & \\ & \end{bmatrix}$$

Eigenvalues: $\lambda = , ,$ (all distinct)

Eigenvectors:

- $\lambda = :$ $(, ,)$
- $\lambda = :$ $(, ,)$
- $\lambda = :$ $(, ,)$

$\boxed{\text{Diagonalizable!}}$ Zero eigenvalue is fine — matrix is singular but diagonalizable.

Example 6: Another Non-Diagonalizable Matrix

$$= \begin{bmatrix} \end{bmatrix}$$

Eigenvalue: $\lambda =$ (algebraic multiplicity $= 3$)

$- \; = \begin{bmatrix} \end{bmatrix}$ Equation: $\; = \;$

Eigenvectors: $\begin{bmatrix} \end{bmatrix}$ with , free

Two independent eigenvectors: $(,,)$ and $(,,)$

Geometric multiplicity $= 2$ ¡ 3 NOT diagonalizable!

Summary: Diagonalizability in $\mathbb{R}^3$

Case	**Diagonalizable?**
3 distinct real eigenvalues	Always
2 distinct (one repeated) with geo mult = alg mult	Yes
2 distinct (one repeated) with geo mult ¡ alg mult	No
1 eigenvalue (mult 3) with geo mult = 3	Yes (scalar matrix)
1 eigenvalue (mult 3) with geo mult = 2	No
1 eigenvalue (mult 3) with geo mult = 1	No

Diagonalizable Enough independent eigenvectors!

Step-by-Step Diagonalization Process

1. Find eigenvalues: $\det(A - \lambda I) = 0$
2. For each eigenvalue λ, find $\dim(\text{eigenspace}) = n - \text{rank}(A - \lambda I)$
3. Check: $\dim(\text{eigenspace}) =$ algebraic multiplicity for all λ
4. If yes, find n independent eigenvectors
5. Form P with eigenvectors as columns
6. Form D with eigenvalues on diagonal (matching order in P)
7. Verify: $AP = PD$

If any geometric multiplicity ¡ algebraic multiplicity $\rightarrow$ STOP! Not diagonalizable.

Practice Problems

1. Diagonalize $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ if possible.
2. Is $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ diagonalizable? Why or why not?
3. Find P and D for $A = \begin{bmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$ (eigenvalues: 10, 1, 1).
4. Determine if $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is diagonalizable.
5. True/False: Every matrix with 3 distinct eigenvalues is diagonalizable. Explain.

Remember: Distinct eigenvalues Always diagonalizable!